

Artificial Economies for Learning Agents: Testing Whether Institutions Stay Robust Across Learning Architectures

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Draft v0.3 – July 27, 2026

Abstract

This paper introduces a code-first experimental platform for studying whether economic mechanisms remain robust when the strategic agents inside them are adaptive learners rather than equilibrium solvers. The platform implements a shared `World/Agent/Institution` interface, repeated multiseed evaluation, exploitability tests, and interchangeable mind classes ranging from random and tabular Q-learning agents to PyTorch DQN, PPO, decorrelated independent-DQN, and centralized-critic agents. The first validated environment is a repeated two-firm pricing arena with six regulatory mechanisms. In this world, price caps robustly reduce exploitability across mind classes, but their effect on collusion is metric- and architecture-dependent. A new paired-seed audit shows that DQN’s positive profit-under-cap result is not visible in 1000-step traces; it emerges with training budget, becomes mostly positive by 10,000 steps, and is positive in the 40,000-step full table. The second environment, Resource Island, extends the same interface to a spatial inventory economy and reveals a different lesson: institution conclusions are only meaningful when the world actually exercises the institutional channel. Early Resource Island versions mechanically under-tested property rights and trade price controls; a hardened v1 now creates contested resources, unequal trades, and specialization pressure. Three additional environments extend the suite: Auction House provides a benchmarked auction-design testbed, Public Goods exposes common-pool extraction and contribution incentives, and Labor Market tests learned reporting under deferred acceptance. The current contribution is both empirical and methodological: it reports initial evidence that institutional robustness changes across learning architectures, and it documents the validation discipline needed to distinguish real economic findings from inactive-mechanism or short-horizon artifacts.

1 Introduction

Economic mechanisms are usually evaluated against a model of strategic behavior: equilibrium prices, truthful bidding, efficient allocation, or some other theoretical benchmark. That evaluation is only as good as the behavioral model behind it. This paper sets out to find out what happens when institutions designed around classical economic assumptions are placed in front of learning agents; which guarantees survive, which break, and does the break depend on the learning architecture?

We answer this with five small, controlled economic worlds, a pricing duopoly (Pricing Arena), a scarce-resource trade economy (Resource Island), a sealed-bid auction (Auction House), a common-pool resource game (Public Goods), and a two-sided labor match (Labor Market), each paired with

a classical benchmark and a shared ladder of learning architectures: random behavior, tabular Q-learning, deep Q-networks, proximal policy optimization, independent multi-agent Q-learning, and centralized-critic training. These architectures are not a scale from weaker to stronger. The question at stake in each world is not which architecture performs best, but which of the institution’s guarantees hold up as the behavioral assumptions behind the benchmark are changed.

Across the five worlds, the same failure modes recur in different forms: an institution can look effective on one metric while a different strategic channel stays open; an apparent effect can depend on training horizon and reverse sign with more training budget; a learning architecture can fail to recover a benchmark that classical theory says is optimal; and an institution can raise a headline welfare number while quietly eroding the property, stability, sustainability, truthfulness, that the institution actually exists to protect. The next section sets out, world by world, the classical benchmark each environment owes and what claim boundary follows from it, while sections 4 through 8 report where each of these failure modes appears, and where the evidence is instead null or still provisional.

2 Related Work

This paper connects five economic literatures to a shared reinforcement-learning testbed. Each world below is built to test one specific classical prediction. The Pricing Arena is motivated by work on algorithmic collusion: repeated-game Q-learning agents can learn to sustain prices above the competitive level, without any explicit communication between them [Calvano et al., 2020, 2021]. The exploitability check used in this paper follows the same concern one step further. A policy’s average performance across a run does not show whether that policy is robust to a deviating opponent, so this paper measures both. Auction House is based on classical auction theory. In a second-price auction, bidding one’s true value is a dominant strategy, meaning it is optimal regardless of what other bidders do [Vickrey, 1961]. In a first-price auction, by contrast, the optimal strategy is to bid below one’s true value (shading), and reserve prices create a known tradeoff between the seller’s expected revenue and the efficiency of the final allocation [Myerson, 1981, Milgrom, 2004].

Public Goods and Resource Island are both based on common-pool resource and public-goods theory. The standard prediction in this literature is that, absent some form of monitoring, reputation, or locally appropriate governance, a shared resource will be overused or a public good will be underfunded, because each agent’s private incentive does not account for the cost imposed on everyone else [Samuelson, 1954, Ostrom, 1990]. Labor Market is based on two-sided matching theory. Under the worker-proposing deferred acceptance algorithm, the resulting match is guaranteed to be stable (no worker and employer who are not matched to each other would both prefer to be), and truthful reporting of preferences is guaranteed to be optimal for the proposing side [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. These are proven properties of the mechanism, not empirical tendencies, which is what makes them useful as benchmarks.

The learner suite is built from five standard reinforcement learning methods: tabular Q-learning, deep Q-networks (DQN) [Sutton and Barto, 2018, Mnih et al., 2015], proximal policy optimization (PPO) [Schulman et al., 2017], independent Q-learning [Tan, 1993], and centralized-critic training [Lowe et al., 2017]. What is different here is that the suite is used to test assumptions, not to compare performance. For example, second-price truthfulness and deferred-acceptance strategy-proofness are proofs about a single, fixed learner responding to a fixed mechanism; independent Q-learning and centralized-critic training instead put multiple agents in motion at the same time, which is not the setting those proofs were written for.

3 Theory Obligations

Every world in this paper owes something to the theory it is built on: a specific prediction it must be checked against, a specific set of measurements that follow from that prediction, and a specific boundary on what the resulting evidence can support. This section states that obligation for each world before any learned-agent result is discussed. The obligation exists because passing tests and producing a complete set of output files is not the same thing as having tested the institution a world claims to test.

The five worlds do not owe the same kind of proof. Grouping them by the strength of their classical prediction gives three tiers: proven mechanism properties, where a learned agent’s behavior can be checked directly against a known correct answer; bracketed or reference-point theory, where exact values exist but no single predicted outcome for the object under test; and activation, where no external benchmark exists at all and the obligation is instead to show the institution was ever placed under real economic pressure.

Auction House and Labor Market belong to the first tier. In Auction House, the classical predictions are exact: truthful bidding in a second-price auction, bid shading in a first-price auction, and a known revenue-efficiency tradeoff under a reserve price. Because the correct behavior is known, this world measures distance from it directly, as revenue, allocative efficiency, bidder surplus, ex-post regret, over- and underbidding, shading distance, and the shape of the learned bid curve against the truthful or shaded reference, rather than raw payoff alone. The resulting claim is bounded to benchmark deviation under different learners; convergence to auction equilibrium is not asserted. In Labor Market, a lower truthful-report rate under some learner is not by itself evidence of profitable manipulation, given the proposing side’s strategy-proofness guarantee. Match rate, blocking-pair stability, truthful-report rate, and welfare are therefore reported together, since a learner can trade one against another, and any claim about manipulation is bounded to the worker-proposing, canonical version of the mechanism implemented here.

Pricing Arena and Public Goods belong to the second tier. Pricing Arena has two exact static prices, the Nash price and the joint-profit price, but no closed-form prediction for behavior in the repeated game itself, only a range bounded by the two. A collusion claim in this world is therefore only as credible as the decomposition behind it: price, profit, quantity, welfare, and exploitability are reported together and checked against each other, and the resulting claim is limited to finite-run, regulatory evidence in a stylized duopoly, not a theorem or a real-market antitrust result. Public Goods is bracketed rather than pointed: theory predicts underprovision when private incentives diverge from the social good, with free-riding as a lower bound and the social optimum as an upper one, and nothing more specific in between. The obligation this creates is to keep the welfare number honest against the underlying resource state, since an institution can raise measured welfare through reward accounting without changing the incentive to extract or contribute at all. Contribution, extraction, sustainability, inequality, and tax revenue are therefore reported as separate quantities from welfare, so that an accounting effect is not mistaken for a real one.

Resource Island belongs to the third tier alone. It carries no external closed-form benchmark, so its obligation is not how close a learned agent gets to a known correct value, but whether the institution being measured was ever placed under real economic pressure in the first place. A property-rights rule is not being tested if no non-owner ever has reason to access a claimed resource; a trade price control is not being tested if every trade happens to be one-for-one already. Survival, sustainability, trade counts, property pressure, specialization, and institution-block counters exist to answer that prior question before any welfare number from this world is trusted. This is not a weaker test than the other four worlds run; it is a test of a condition the other four simply assume already holds, and Section 5 reports a case where that assumption failed and had to be corrected

before the world’s results meant anything.

4 Platform

4.1 Interfaces

The codebase is organized around three abstractions:

- A **World** defines state transitions, observations, metrics, and rendering state.
- An **Agent** or mind maps observations to actions and updates from rewards and next observations.
- An **Institution** modifies rules, rewards, constraints, or information inside a world.

This separation is not cosmetic. It makes it possible to ask whether a result is about a world, an institution, or a learner. It also creates a regression test: if a new world requires rewriting the shared learning interface, then the platform abstraction has failed.

4.2 Validation Discipline

The platform treats a result as reportable only after four checks. First, the mechanics must be unit-tested: allocation, movement, payment, trade, collision, or reward rules must match hand-computed cases. Second, where theory provides a known answer, the implementation must recover or bracket that answer before learned-policy output is interpreted. The current known-answer runner reports 19/19 passing checks across Pricing Arena N-firm static Nash references, Auction House truthfulness and bid-shading cases, Public Goods free-rider and cooperative brackets, Labor Market deferred-acceptance benchmark cases, and Resource Island gather bounds. Third, the experiment must be replicated over seeds, with per-seed and aggregate CSV outputs. Fourth, the institution must actually activate. An institution that never faces the behavior it is meant to govern produces no evidence about that institution, only about the world design. Section 5.2 gives a concrete case.

The second new validation layer is mechanism tracing. The trace runner produces per-step rows and algebraic decomposition rows for all five worlds and all thesis-facing minds. These traces do not replace full-run evidence; they show the channel behind a claim and can expose contradictions between short traces and full-run results. Section 4.3 gives a concrete case.

4.3 Learner Axis

The learner suite is read as an assumption-violation axis rather than a capability ranking. The table below lists the assumption each learner changes.

Learner	Assumption Changed	Interpretation Role
Random	No strategic learning.	Baseline scale for world rewards and metrics.
Tabular Q-learning	Exact finite table, no function approximation [Sutton and Barto, 2018].	Closest discrete learner to the textbook single-agent control.
DQN	Function approximation, replay, target network, and approximate value representation [Mnih et al., 2015].	Tests whether approximation changes institutional response.
PPO	On-policy stochastic policy optimization with clipped updates [Schulman et al., 2017].	Tests whether policy-gradient learning finds different strategic channels.
Independent-DQN	Separate learners interacting in a non-stationary multi-agent environment [Tan, 1993].	Tests whether independent learning changes stability and coordination.
Centralized critic	Centralized training signal for decentralized actors [Lowe et al., 2017].	Tests whether extra training information helps or destabilizes each institution.

The DQN implementation uses replay, a target network, Huber TD loss, and a DeepMind-style centered RMSProp optimizer [Mnih et al., 2015]. PPO uses categorical actions, GAE, clipped policy loss, value loss, entropy regularization, and minibatch epochs [Schulman et al., 2017].

An important implementation issue was discovered and fixed during the project: the original independent-DQN condition behaved like a DQN alias in several tables [Mnih et al., 2015, Tan, 1993]. The corrected implementation now uses separate per-agent learners, decorrelated child random streams, separate replay and exploration randomness, and scoped PyTorch initialization. Fixed independent-DQN replacement rows are now merged for Pricing Arena and Resource Island, and the same corrected condition is used in the completed Auction House, Public Goods, and Labor Market P.6 full learner-suite runs.

5 Environment 1: Pricing Arena

5.1 World

Pricing Arena is a repeated two-firm pricing market. Each firm chooses a price from a discrete grid every round. Consumers divide between the two firms according to a noisy logit demand model, so a firm’s realized quantity depends on both its own price and its rival’s. Profit is price minus cost, times realized quantity, and learning agents update from repeated play against each other. Six mechanisms are implemented on top of this base game: no regulation, a price cap, a tax on high prices, random audits, a direct anti-collusion penalty, and demand shocks.

5.2 Metrics

The world reports average price, total profit, consumer surplus, welfare, price dispersion, exploitability, and two collusion indexes. The first is a price-normalized proxy used since early versions of the project. The second is a profit-normalized index, used because it maps directly onto the collusion measure common in the algorithmic-pricing literature:

$$CI_{profit} = \frac{\pi_{observed} - \pi_{Nash}}{\pi_{monopoly} - \pi_{Nash}}. \quad (1)$$

Both are reported together, since the two can move in different directions, and that divergence is itself informative about which channel a mechanism is actually affecting.

5.3 Known-Answer Validation

Before any learned result is interpreted, the static Nash best response is checked directly against the world’s own demand and cost parameters, for $n = 2, 3, 4$, and 5 firms. All four checks pass. This does not validate anything about learned behavior; it validates that the reference point learned behavior is measured against is not arbitrary.

5.4 Baseline and Main Result

Table 1 shows the unregulated baseline for each learner. Q-learning and the centralized critic sit at the higher-price, higher-collusion end of the range; DQN, PPO, and independent-DQN sit lower, with lower exploitability.

Mind	Avg. Price	Profit	Welfare	Collusion Index	Exploitability
Q-learning	5.487	365.3	810.4	0.541	95.7
DQN	3.611	236.3	832.8	0.202	20.3
PPO	3.681	260.0	839.3	0.215	14.2
Independent-DQN	4.078	269.9	827.9	0.285	26.6
Centralized critic	5.925	288.6	791.6	0.618	147.5

Table 1: Pricing Arena, no-regulation baseline, by learner.

Table 2 shows the change under a price cap, relative to that baseline, for every learner. The exploitability column is the clearest result in this world: it falls for every learner, without exception. The profit column is not: it falls for Q-learning and random play, in line with the conventional expectation that a cap disciplines pricing behavior, but it rises for DQN, independent-DQN, and the centralized critic.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	−0.759	−22.916	+19.567	−55.366
Random	−1.185	−16.028	+34.779	−31.775
DQN	+0.388	+55.358	+6.294	−11.619
PPO	−0.042	−6.558	−1.198	−5.704
Independent-DQN	−0.288	+3.093	+12.183	−14.871
Centralized critic	−1.337	+10.347	+28.668	−98.274

Table 2: Price cap versus no regulation, change by learner, n=20 full run.

The pattern is a guardrail that holds on the axis it was built for and does not hold on an adjacent one. A price cap constrains the price level directly, and every learner’s exploitability falls as a result, since a capped price cannot be pushed into the range a deviating opponent could exploit most severely. But the cap does not constrain profit directly, only through price, and for three of

the five learners the resulting change in realized quantity and market structure more than offsets the lower price. Q-learning behaves close to the textbook intuition. DQN, independent-DQN, and the centralized critic do not.

5.5 The Training-Budget Correction

The DQN result above was not visible in the first version of this analysis. A short, 1000-step mechanism trace, run to check the channel behind the headline number, showed DQN profit falling under the cap for every seed tested, the opposite of Table 2. This contradiction was not resolved by picking the number that fit the argument. It was resolved by running the same paired-seed comparison at a range of training budgets, from the short trace up to the full 40,000-step run the headline table is drawn from.

Source	Steps	Mean Profit Δ	Positive Seed Share	Mean Quantity Δ
fresh audit	1,000	-11.711	0.10	3.220
fresh audit	5,000	+1.730	0.50	4.102
fresh audit	10,000	+35.448	0.80	3.540
full table	40,000	+55.358	0.85	0.226

Table 3: DQN price-cap profit delta by training budget. Values are paired seed deltas, price cap minus no regulation.

The sign flips between 1,000 and 5,000 steps and is positive with a clear majority of seeds by 10,000. At 40,000 steps, the profit effect is statistically positive ($\Delta\pi = 55.358$, 95% CI [4.859, 105.857], positive in 85% of seeds), but the mean quantity change at that budget is small, unlike the 5,000 and 10,000-step rows where quantity moves substantially. This rules out the simplest explanation, that DQN is exploiting a fixed quantity-margin loophole from the start; the effect is not present early and does not have a single stable channel behind it at every budget. The safer claim is that the price cap becomes a focal constraint some DQN runs learn to use to their advantage only after enough training, not a channel visible or exploitable from the beginning. A short trace is not sufficient evidence either way for a result like this one; the training-budget ladder is what makes the finding reportable.

5.6 The Other Four Mechanisms

Price cap is the only mechanism in this world with a result this clean, and that asymmetry is itself worth reporting rather than hiding. The remaining four are correctly implemented and pass the same unit and parity checks as price cap, but three of the four are weak, inert, or inconsistent enough across learners that no single clear claim follows from them at the currently tested parameters.

The high-price tax subtracts a quantity-weighted penalty from reward above a price threshold. Its effect is small in every direction and every learner: Q-learning’s exploitability falls modestly, DQN is essentially unaffected, and the centralized critic moves in the wrong direction on price and welfare. At the tested tax rate, the penalty is too small, or too easy to avoid, to reliably change pricing behavior.

Random audits apply a stochastic penalty when average price is high. The result here is worse than merely weak: for Q-learning, exploitability rises under audits rather than falling, and for the centralized critic, profit rises sharply while exploitability falls only partially. An enforcement

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	-0.078	+2.2	-7.6
DQN	+0.006	+0.6	+1.2
PPO	-0.023	-3.6	-5.2
Independent-DQN	+0.001	+0.2	+2.5
Centralized critic	+0.088	+4.2	-1.2

Table 4: High-price tax versus no regulation, change by learner.

mechanism that increases exploitability for one learner while partially working for another is not a guardrail; at these parameters it is closer to noise with an inconsistent sign.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	+0.114	+11.1	+32.0
DQN	+0.035	+4.3	-0.5
PPO	-0.006	-1.0	-2.7
Independent-DQN	+0.002	+0.2	+1.7
Centralized critic	+0.125	+49.1	-21.7

Table 5: Random audit versus no regulation, change by learner.

The direct anti-collusion penalty punishes prices that are both high and close together, the most explicit mechanism in the suite and, on paper, the one closest to actual antitrust enforcement logic. It still mostly fails to bind: four of the five learners show negligible or worsened exploitability, and only PPO shows a mild, coherent improvement across price, profit, and exploitability together. A rule aimed directly at the behavior of interest is not automatically more effective than the blunter price cap.

Mind	Price Δ	Profit Δ	Exploitability Δ
Q-learning	-0.048	-2.1	+31.9
DQN	+0.005	+0.5	0.0
PPO	-0.067	-7.4	-5.5
Independent-DQN	+0.001	+0.2	0.0
Centralized critic	-0.013	+0.1	+0.5

Table 6: Anti-collusion penalty versus no regulation, change by learner.

Demand shocks perturb market size with a lognormal multiplier each round, testing robustness to environmental volatility rather than acting as a policy lever. Results are small and mixed in direction across learners, with the centralized critic the only one meaningfully affected, and negatively. This mechanism is best read as a stress test the platform passes mechanically, not as a candidate institutional finding.

None of the four should be read as a code defect. All four pass the same mechanics and parity tests as price cap. The difference is economic activation, not implementation: at the parameters currently tested, only price cap creates pressure strong enough to produce a consistent, cross-learner, interpretable signal. Table 8 summarizes the distinction.

Mind	Price Δ	Profit Δ	Welfare Δ	Exploitability Δ
Q-learning	+0.014	+4.0	+0.3	+11.4
DQN	-0.002	+1.3	+1.8	+1.3
PPO	-0.010	-0.3	+1.4	-6.0
Independent-DQN	+0.001	+0.6	+1.2	+1.8
Centralized critic	-0.050	-9.2	-12.0	+2.7

Table 7: Demand shock versus no regulation, change by learner.

Mechanism	Implementation	Interpretability
Price cap	Passes all checks	Strong, consistent, cross-learner
High-price tax	Passes all checks	Weak at tested rate
Random audit	Passes all checks	Mixed, sign-inconsistent
Anti-collusion penalty	Passes all checks	Mostly inert, brittle
Demand shock	Passes all checks	Robustness check, not a policy result

Table 8: Mechanism status, implementation versus interpretability, Pricing Arena.

5.7 Summary

Pricing Arena’s clearest result is a guardrail that succeeds on the axis it directly targets and fails to hold as a guarantee on an adjacent one: price caps lower exploitability for every learner tested, but their effect on profit and collusion is learner-dependent, with the effect for DQN emerging only after sufficient training rather than being present from the start. The remaining four mechanisms are correctly implemented but, at current parameters, mostly too weak or inconsistent to support a claim beyond their own numbers. Whether the price-cap result and the weakness of the other four persist as the number of firms increases beyond two is addressed in the cross-world synthesis section.

6 Environment 2: Resource Island

6.1 World

Resource Island is a bounded-grid gather/trade economy. Agents move, gather food and wood, maintain energy, accumulate inventories, and can die from starvation. The world contains property rights, redistribution, trade price controls, and reputation institutions. It was built to test whether the platform abstractions work outside a two-firm pricing game.

6.2 What v0 Taught

The first full Resource Island runs were mechanically valid but economically under-activated. In the initial full run, successful trade was sparse and some institutions did not bind. Follow-up replay showed that property rights almost never faced behavioral pressure: even when claims existed, non-owner opportunities to gather from claimed resource cells were extremely rare. Similarly, trade price controls could not bind because the original trade protocol only allowed one-for-one exchange.

This is an important negative result for the platform: a world can be runnable, tested, and statistically replicated while still failing to exercise the institution it is supposed to study.

6.3 v1 Hardening

Resource Island v1 adds four changes, motivated by common-pool-governance and public-goods work on monitoring, local fit, incentives, and repeated interaction [Ostrom, 1990, Fehr and Gächter, 2000, Isaac et al., 1994]:

1. contested resource layouts, so multiple agents have reason to access the same cells;
2. unequal trade offers, so price controls can actually bind;
3. specialization pressure through heterogeneous resource preferences and starting inventories;
4. activation diagnostics for trade attempts, blocked trades, property opportunities, property violations, and institution blocks.

The v1 validation run verified that the economic channels were active, and the subsequent n=20 full run preserves that activation. Table 9 shows the medium validation values used as the pre-full-run gate.

Institution	Welfare	Survival	Trades	Trade Blocks	Property Opps.
none	1.095	0.933	6.034	0.000	0.000
property rights	1.078	0.923	5.790	0.000	17.734
trade price controls	0.875	0.898	0.000	5.779	0.000
reputation	1.313	0.927	6.867	0.000	0.000

Table 9: Resource Island v1 validation results.

The interpretation is straightforward. Price controls now bind because the world permits unequal trades. Reputation increases successful trade and welfare in the validation setting. Property-right opportunities are present, although the welfare effect is small in this medium run. The full n=20 v1 run confirms the activation pattern: successful trade appears under no institution, price controls bind and suppress trade, property-right opportunities are measured, and reputation produces the strongest welfare gains in this pressure setup. The strict-local ablation confirms that this remains partly a spatial-friction story: reducing trade from all-island matching to radius-one local matching cuts baseline trade from 5.325 to 1.900 and reputation trade from 6.667 to 1.927, while property opportunities and price-control blocks remain positive.

6.4 Cross-Mind Resource Island

The v0 Resource Island Phase 3 full table initially gave a different pattern from Pricing Arena: neural/MARL minds improved survival and welfare but did not learn successful trade. After the v1 pressure changes, that is no longer the main Resource Island conclusion. Table 10 shows the no-institution row from the v1 full learner-suite run.

The updated result is more useful. PPO and centralized-critic discover the trade channel under pressure, while DQN and fixed independent-DQN remain more conservative and trade much less. Under trade price controls, successful trade falls to zero across the learner suite because unequal exchange is blocked. Resource Island therefore contributes a different kind of learner-architecture effect from Pricing Arena: stronger learners do not simply exploit or avoid an institution monotonically; their ability to activate the intended market channel depends on the learning rule and on whether the environment creates enough pressure for exchange.

Mind	Welfare	Trades
Q-learning	1.107	5.325
DQN	1.061	0.640
PPO	1.254	8.892
Independent DQN	1.063	1.730
Centralized critic	1.248	8.697

Table 10: Resource Island v1 full learner-suite run, no-institution condition.

7 Environment 3: Auction House

7.1 World

Auction House is a single-item sealed-bid auction world with independent private values, discrete bids, deterministic tie-breaking, first-price and second-price payment rules, and optional reserve prices [Vickrey, 1961, Myerson, 1981, Milgrom, 2004]. Each bidder observes its own valuation bin and chooses a bid. The world reports seller revenue, bidder surplus, total welfare, allocative efficiency, regret, overbidding, underbidding, and distance from benchmark bid functions.

7.2 Benchmarks

The second-price benchmark is truthful bidding [Vickrey, 1961]. The first-price benchmark is bid shading under symmetric independent private values, with discrete-grid regret checks used where the continuous benchmark is too coarse [Milgrom, 2004, Dütting et al., 2019, Banchio and Skrzypacz, 2022]. Reserve-price variants evaluate the revenue-efficiency tradeoff against the no-reserve benchmark [Vickrey, 1961, Myerson, 1981, Milgrom, 2004].

7.3 Validation

The Auction House validation run is economically coherent but not yet final evidence. Second-price learning moves toward the truthful benchmark, first-price learning shows underbidding and bid shading, and reserve prices increase revenue while changing allocative efficiency. Table 11 summarizes the current validation run.

Scenario	Revenue	Welfare	Efficiency	Underbid Rate
second price	3.373	6.216	0.747	0.381
first price	3.481	6.331	0.768	0.774
second price + reserve	4.031	6.098	0.814	0.366

Table 11: Auction House validation results.

7.4 Cross-Mind Auction Diagnostics

The full P.6 auction learner-suite run confirms that the auction world should be read against benchmark deviations, not raw reward alone. In second-price auctions, PPO has the lowest ex-post regret among the tested learning minds (0.187) and higher allocative efficiency than Q-learning (0.762 versus 0.734). DQN and fixed independent-DQN earn higher seller revenue but with lower

allocative efficiency. The centralized-critic scaffold performs poorly in this asymmetric bidding setting, with low revenue and high regret.

First-price auctions show the expected underbidding/shading channel across the non-centralized learners [Milgrom, 2004]. Q-learning, DQN, PPO, and fixed independent-DQN all underbid frequently, with first-price underbid rates between 0.716 and 0.814 [Milgrom, 2004]. This makes Auction House theory-anchored: the economic question is how each learner deviates from truthful second-price bidding or shaded first-price bidding, not whether the environment merely produces positive payoffs [Vickrey, 1961, Milgrom, 2004, Dütting et al., 2019].

8 Environment 4: Public Goods

8.1 World

Public Goods is a repeated common-pool resource environment, following the standard tension between private extraction and shared surplus [Samuelson, 1954, Hardin, 1968]. Agents choose between contributing to the pool, extracting from it, or remaining inactive. Extraction pays private reward but depletes the shared stock; contribution costs the contributor but helps sustain the pool. When requested extraction exceeds available stock, extraction is rationed proportionally. Deterministic regeneration makes the sustainability channel easy to audit.

The implemented institutions are no intervention, a penalty schedule, contribution matching, reputation rewards, information restriction, and a generic tax-and-redistribution schedule. The main metrics are welfare, sustainability, total contribution, extraction relative to regeneration, inequality, collapse diagnostics, and tax revenue where applicable.

8.2 Q-Learning Full Run

The $n=20$ tabular Q-learning full run confirms the intended commons pressure. In the baseline, agents mostly extract and contribute little: sustainability is 0.0892 and total contribution is 0.1418. Contribution matching improves both welfare and sustainability in this setting, reaching welfare 2.0808 and sustainability 0.1048. Reputation is a strong reward-shaping intervention, with welfare 9.5750, while the tax schedule mainly changes redistribution/accounting at the tested rates.

The effect validator classifies the tax schedule as mostly reward/accounting under this configuration, while penalty, contribution matching, reputation, and information restriction change state metrics relative to baseline. This is a diagnostic classification, not yet a final welfare ranking.

8.3 Cross-Mind Status

The full Public Goods learner suite shows that stronger function approximation or centralized training does not automatically solve the commons problem. Under no institution, Q-learning reaches welfare 1.815, sustainability 0.089, and contribution 0.142. DQN and fixed independent-DQN are close but contribute less; PPO and centralized-critic contribute almost nothing and sit near the lower sustainability regime.

Contribution matching is the clearest state-changing intervention in the full learner-suite run. It improves Q-learning sustainability from 0.089 to 0.105 and fixed independent-DQN sustainability from 0.085 to 0.121, with a larger increase in contribution for independent-DQN. It does not rescue PPO in this configuration. The economic reading is therefore conditional: the institution can improve the state variables when the learner explores contribution, but a more capable policy class can still converge to low-contribution behavior.

9 Environment 5: Labor Market

9.1 World

Labor Market is a two-sided matching environment. Workers are learning agents; employers have fixed preferences. Each worker reports a top choice, the world constructs reported preferences, and matching is resolved using deferred acceptance [Gale and Shapley, 1962, Roth and Sotomayor, 1990]. The world reports match rate, total welfare, truthful-report rate, stability, blocking-pair diagnostics, and manipulation-gain diagnostics.

This world is intentionally asymmetric. Unlike Pricing Arena, Resource Island, Public Goods, and Auction House, not every economic actor is a learning agent. That asymmetry is the main interface test for the learning-architecture suite: DQN, PPO, independent-DQN, and centralized-critic operate only over the worker side while employers remain fixed preference holders.

9.2 Q-Learning Full Run

The $n=20$ tabular Q-learning full run is substantially more stable than the short smoke run. Match rate remains 1.0, while `stability_mean=0.9508`, `truthful_report_rate_mean=0.7859`, and `total_welfare_mean=3.6393`. The report-top action space creates truthfulness variation, but worker-proposing deferred acceptance still produces mostly stable matchings under the current random-preference setup.

Fixed benchmark cases clarify the interpretation. Under worker-proposing deferred acceptance, workers should not have profitable report deviations in the canonical strategy-proof setting [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. Future manipulation tests therefore need to target the non-proposing side, change the mechanism, or add information and commitment frictions rather than treating worker misreports under standard deferred acceptance as the main expected failure mode.

9.3 Cross-Mind Status

The full Labor Market learner-suite run keeps match rate at 1.0 for every mind, so the relevant margins are stability and truthful-report behavior. DQN slightly improves stability relative to Q-learning (0.975 versus 0.951) while reporting truthfully less often. PPO and fixed independent-DQN preserve high stability but also lower truthfulness. The centralized-critic scaffold has the weakest matching behavior in this asymmetric setup, with stability 0.749 and truthful-report rate 0.433.

Economically, this is consistent with the benchmark warning: worker-proposing deferred acceptance is strategy-proof for workers in the canonical setting, so lower truthful-report rates are not automatically evidence of profitable manipulation [Gale and Shapley, 1962, Roth and Sotomayor, 1990, Roth, 1984]. The result is better framed as a learned-reporting diagnostic: the mechanism keeps matches mostly stable for most learners, while the centralized-critic architecture is brittle in this asymmetric environment.

10 Cross-World Synthesis

The emerging thesis is not that one institution always works, or that the agent classes form a universal intelligence ranking. The stronger claim is more specific: institutional robustness must be evaluated jointly with the learning architecture, the training horizon, and activation diagnostics.

Tabular value learning, function approximation, on-policy policy gradients, decorrelated independent learners, and centralized critics violate different assumptions of the classical benchmark story. The comparisons are therefore best read as a learner-architecture stress test, not as a scalar capability ranking.

The Pricing Arena result shows architecture-sensitive institutional effects: price caps reduce exploitability broadly, but DQN’s profit response flips sign with training budget and is positive in the full paired audit. Resource Island shows the complementary failure mode: if the world design does not put an institution under pressure, then the absence of an effect is not a mechanism result. Auction House adds theory-anchored bidding benchmarks [Vickrey, 1961, Myerson, 1981]. Public Goods adds a sustainability and commons-collapse channel. Labor Market adds a matching-stability and truthfulness channel.

The five-world synthesis is now built from full $n=20$ learner-suite outputs for Pricing Arena, Resource Island v1, Auction House, Public Goods, and Labor Market. The protocol-comparability report passes for all five worlds, and the publication inference table includes all worlds under the same confirmatory bookkeeping. The claim-audit suite adds a second pass over those outputs: paired seed effects, benchmark gaps, metric-conflict warnings, activation confirmations, and trace/full-run sign agreement where paired traces exist. In its current run, the audit produces 1,025 paired or benchmark rows and 31 metric-conflict or activation rows. This audit layer prevents the cross-world synthesis from becoming a table of averages without claim boundaries.

The main synthesis claim is qualitative but concrete. Learning architecture changes institutional outcomes in every world, but not through one universal direction. In Pricing Arena, DQN’s price-cap profit increase is a long-training, seed-heterogeneous effect rather than a short-run channel. In Resource Island v1, PPO and centralized-critic activate trade strongly while DQN-family learners remain more conservative [Mnih et al., 2015, Schulman et al., 2017]. In Auction House, the relevant question is distance from truthful, shaded, or reserve-price benchmarks rather than raw reward [Vickrey, 1961, Myerson, 1981, Milgrom, 2004]. In Public Goods, the relevant state variables are contribution, extraction, and sustainability, while some tax effects are accounting effects rather than state changes. In Labor Market, deferred acceptance produces high match rates and mostly stable outcomes, so future manipulation claims must target the right side of the market or a different matching institution.

11 Limitations

Several limitations are deliberate. The environments are small and stylized. They are designed for auditability rather than realism. The deep-RL agents are structured-observation MLP agents rather than large-scale policies. The Resource Island trade protocol is still simple, and the current v1 pressure configuration is an activation test rather than a realistic market. Auction House currently has benchmarked sealed-bid full runs and smoke-tested clock and information variants. Public Goods is still a stylized common-pool model rather than an empirical commons. Labor Market currently uses worker-side learning under deferred acceptance, so it tests one asymmetric matching channel rather than matching-market design generally.

The most important limitation is interpretive. A clean CSV does not imply a clean economic result. Every institution must be checked for activation. Every learner-architecture comparison must be checked for implementation artifacts. The independent-DQN alias issue is a concrete example: the table was statistically valid as an output file, but not valid as evidence for a distinct mind until the implementation was corrected and rerun.

The Pricing Arena price-cap audit adds a second interpretive warning. A mechanism trace

can reveal the local channel but cannot replace a replicated training-budget audit. The 1000-step trace correctly showed early profit losses under the cap; the 40,000-step table correctly showed a positive DQN cap-minus-none profit delta. The result is reportable only because the contradiction was resolved through paired seed deltas and a budget ladder.

12 Reproducibility

The repository records validation status in `ROADMAP_STATUS.md`. Core outputs are stored under `outputs/`. Major runners include:

- `run_multiseed.py` for Pricing Arena replicated runs;
- `run_exploitability.py` for frozen-policy adversarial evaluation;
- `run_phase3_full.py` and `validate_phase3_full.py` for full Phase 3 Pricing Arena checks;
- `run_known_answer_sanity_checks.py` for theorem/bracketing checks before learned-policy interpretation;
- `run_mechanism_traces.py` for per-step mechanism traces and decomposition tables across worlds;
- `run_pricing_quantity_margin_audit.py` for the Pricing Arena price-cap budget-ladder contradiction audit;
- `run_claim_audit_suite.py` for cross-world paired-effect, benchmark-gap, metric-conflict, activation, and trace-alignment audits;
- `run_resource_island_smoke.py` for Resource Island smoke, validation, and full runs;
- `run_auction_house_smoke.py` for Auction House runs;
- `run_public_goods_smoke.py` for Public Goods runs;
- `run_labor_market_smoke.py` for Labor Market runs;
- `build_world_mind_comparison.py` and `build_cross_world_synthesis.py` for ladder comparisons and cross-world synthesis.

The test suite currently includes unit, integration, and output-validation coverage for the core abstractions, Pricing Arena, Resource Island, Auction House, Public Goods, Labor Market, and deep-RL/MARL minds.

12.1 Primary Output Artifacts

The main numerical claims in the paper are backed by the following generated artifacts:

- `outputs/known_answer_sanity_checks_current/summary.csv`;
- `outputs/mechanism_traces_local_full/mechanism_decomposition.csv`;
- `outputs/phase3_full/mind_comparison.csv`;

- `outputs/pricing_quantity_margin_audit/dqn_budget_ladder.csv`;
- `outputs/pricing_quantity_margin_audit/existing_full_seed_deltas.csv`;
- `outputs/resource_island_v1_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_strict_radius_full/summary_aggregate.csv`;
- `outputs/resource_island_v1_phase3_full/mind_comparison.csv`;
- `outputs/auction_house_phase3_full/mind_comparison.csv`;
- `outputs/public_goods_phase3_full/mind_comparison.csv`;
- `outputs/labor_market_phase3_full/mind_comparison.csv`;
- `outputs/cross_world_synthesis/synthesis_table.csv`;
- `outputs/claim_audit_suite/claim_audit_report.md`.

13 Conclusion

This draft reports a first version of an artificial-economies platform for testing institutional robustness under learning agents. The mature Pricing Arena result already shows why the question matters: exploitability and collusion can move differently under the same institution, and a headline effect can depend on training budget rather than appearing in short traces. Resource Island, Auction House, Public Goods, and Labor Market extend the project beyond pricing games. Their role is not merely to add more checkboxes, but to test whether the patterns found in Pricing Arena survive when the economic structure changes and when the relevant institutional channel is actually activated.

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